

James Makhoul

Portfolio | LinkedIn | GitHub

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MECHATRONICS ENGINEERING STUDENT

Fourth-year Mechatronics Engineering student with a perfect GPA and Dean's List recognition. Strong interest in control systems, robotics, and brain-computer interfaces (BCI), with hands-on experience in research, embedded systems, and software development. Actively engaged in interdisciplinary projects bridging mechanical, electrical, and computational engineering.

TECHNICAL SKILLS & SOFT SKILLS

- **Engineering Tools:** MATLAB/Simulink, SOLIDWORKS, AutoCAD, ROS 2, CMake, Docker
- **Programming and Development:** C++, Python, Java, Flutter (Dart), JavaScript, HTML, CSS, PostgreSQL, Firebase
- **Machine Learning:** Deep Learning, Computer Vision, Neural Networks
- **Soft Skills:** Problem Solving, Critical Thinking, Adaptability, Time Management, Teamwork

EXPERIENCE

Robotics Intern <i>Inmind.ai</i> • Developed and executed SLAM simulation pipelines in Dockerized ROS 2 environments. • Built and debugged multi-node ROS 2 packages integrating driver, filter, and logger nodes for real-time processing.	July 2026 – Present <i>Mkalles, Lebanon</i>
Engineering Intern <i>Mitsulift</i> • Drafted 2D AutoCAD engineering drawings, including proposal layouts, shop drawings, and installation diagrams. • Collaborated with the sales engineering team on cost estimation and drafting technical project commercial offers. • Performed Quality Control (QC) field inspections on installed elevator systems, auditing mechanical integrity and logging snag reports. • Designed a detailed 3D elevator and counterweight assembly in SOLIDWORKS, configuring mechanical mates to simulate cable motion. • Constructed a functional 2-floor prototype elevator controlled by an Arduino Mega with IR sensors, limit switches, and motor actuators.	Jun 2026 – Aug 2026 <i>Dbayeh, Lebanon</i>
Undergraduate Research <i>Lebanese American University</i> • Conducting research in Brain-Computer Interfaces (BCI), focusing on multi-channel EEG signal preprocessing.	Dec 2025 – Present <i>Byblos, Lebanon</i>

PROJECTS

T.R.A.C.E. (Writing Robotic Arm) • Designed and 3D-printed a 3-DOF robotic arm. • Implemented a Task-Space Impedance Controller for precise trajectory tracking. • Fully simulated on MATLAB using Simscape.	<i>Arduino, MATLAB/Simulink</i>
Nine Lives (Game Development Project) • Developed a multiplayer and singleplayer pixel-art platformer using Flutter and the Flame engine. • Implemented Firebase real-time database for real-time multiplayer gameplay. • Hosted on https://nine-lives-game.web.app/	<i>Flutter (Dart), Flame Game Engine, Firebase</i>
Inverted Pendulum Control and Swingup • Designed and implemented swing-up and stabilization controllers by applying PID, fuzzy logic, LQR, and energy-based control strategies.	<i>MATLAB/Simulink</i>
Explorer Bot • Developed an award-winning glove-controlled robotic system featuring a mechanical grabber, camera, and multiple sensors.	<i>Arduino, myRIO</i>
Roommate Finder • Designed and developed a web platform to match users with compatible roommates based on lifestyle, location, and budget.	<i>Node.js, Express, JavaScript, CSS, HTML</i>

EDUCATION

Lebanese American University <i>Bachelor in Mechatronics Engineering (Dean's Distinguished List GPA: 4.0/4.0)</i>	Byblos, Lebanon Aug 2023 – Present
College Notre Dame de Louaize <i>Lebanese Baccalaureate - General Sciences (High Distinction Final Average: 18.167)</i>	Zouk Mosbeh, Lebanon Sep 2008 – Jun 2023

CERTIFICATIONS & AWARDS

- Certified SolidWorks Associate (CSWA) by Dassault Systèmes
- LAU Engineering Projects Day Winner (Explorer Bot)
- 1st Place in the Lebanese National Mathematics Olympiad by CRDP

VOLUNTEERING

- LAU Robotics Summer Camp 2025: Mentored high school students during hands-on robotics workshops.

LANGUAGES

- **English:** C2 | **French:** C1 | **Arabic:** C1